

Synopsis / Project Proposal

1. Project Title and Group

Field	Details
Project Title	Project Management System
Group Number	Team 14
Course	UE24CS341A - Software Engineering
Development Methodology	Agile

2. Team Profile

Sl. No.	Name	USN	SRN
1	Samarth M H	PES1UG24CS407	PES1202403674
2	Sameeksha Prasanna Kumar	PES1UG24CS410	PES1202400207
3	Sanapureddy Hemanth Kumar Reddy	PES1UG24CS418	PES1202400144
4	Sanjana Guttapally	PES1UG24CS420	PES1202400184

3. Problem Statement

Teams working on projects today rely heavily on spreadsheets, chat messages and verbal updates to track tasks and deadlines. This leads to unclear task ownership, missed deadlines, duplicated work and no real visibility into how a project is actually progressing. Managers struggle to see team workload at a glance, and team members often do not know the current priority or status of their tasks. There is a need for a single system that lets a team plan work, track progress and see project health in real time without depending on scattered tools.

4. Proposed Project Description

Orbit is a web based project management system that allows a team to organize work into workspaces and projects, break work down into tasks and subtasks, plan and execute Agile sprints, track time spent on tasks, and monitor overall project health through a live dashboard. The system supports role based access so that different members of a team, such as an owner, admin or regular member, get access appropriate to their responsibility. Task boards update in real time using WebSockets, so that when one member moves a task or leaves a comment, every other connected member sees the change immediately without refreshing the page. The system is being developed for use by student project teams and small work groups who need a lightweight but capable alternative to tools such as Jira or Trello.

Once built, the system will let a user do the following:

- Create an account, create or join a workspace, and be assigned a role
- Create projects within a workspace and add team members to them
- Create tasks and subtasks, assign them, set due dates and mark dependencies between tasks
- View and manage tasks on a drag and drop Kanban board
- Create a product backlog and organize backlog items into sprints with story point estimates
- Log time spent against a task and compare it with the original estimate

- Comment on tasks, mention teammates, and see a live activity feed
- Receive in app notifications for relevant updates
- View an analytics dashboard showing sprint burndown, team velocity and workload distribution

5. Target Users and Stakeholders

User Class	Description
Team Member	Works on assigned tasks, logs time, updates task status, comments on tasks
Project Manager / Owner	Creates projects, plans sprints, assigns tasks, monitors progress and reports
Administrator	Manages workspace members, roles and overall system configuration

Stakeholder: the course instructor acts as the customer for the purpose of this project, representing a team lead who needs visibility into ongoing work.

6. Objectives

- Reduce the time a team spends manually tracking project status
- Provide real time visibility into task progress across a project
- Enable structured, sprint based planning that follows Agile principles
- Provide accurate time logging and workload visibility for each team member
- Provide actionable analytics on team velocity and project health
- Develop the product incrementally using Agile practices with working demos every sprint

7. Scope

In Scope	Out of Scope
User registration, login and role based access control	Payroll or billing integration
Workspace and project creation and management	Two way sync with external calendars
Task and subtask management with a Kanban board	Native mobile applications
Task dependencies with cycle detection	Enterprise single sign on
Agile backlog and sprint planning with story points	Offline mode
Time logging against tasks	
Real time task board updates using WebSockets	
Comments, mentions and an activity feed	
In app notifications	
Analytics dashboard with burndown and velocity charts	

8. Functional Features

- Workspace and account management with role based access control
- Project and task management with a drag and drop Kanban board
- Agile sprint planning with backlog and story point estimation
- Time tracking and workload management
- Real time collaboration on task boards using WebSockets
- Comments, mentions and a live activity feed
- In app notifications
- Analytics and reporting dashboard

9. Plan of Work and Team Ownership

9.1 Team Wise Functional Ownership

Member	Owned Functionality	Secondary Responsibility
Samarth M H	Auth, Workspaces and Access Control: registration, login, workspace and organization creation, invite links, role based permissions	API design and review
Sameeksha Prasanna Kumar	Project and Task Engine: project and task/subtask CRUD, Kanban board, task dependencies with cycle detection	UI/UX design
Sanjana Guttapally	Agile Sprint and Time Management: backlog, sprint planning, story point estimation, time logging, burndown and velocity charts	Database schema design
Sanapureddy Hemanth Kumar Reddy	Real Time Collaboration, Notifications and Analytics: live board updates using Django Channels/WebSockets, comments, activity feed, notifications, analytics dashboard	Testing and CI

Note: UI design, database work and testing are supporting responsibilities and are not treated as a substitute for owning a functional feature. Every member owns one complete, end to end functionality as required.

9.2 Short Term Plan (Sprint Wise)

Sprint	Planned Output	Lead	Acceptance Evidence
Sprint 1	Project setup, requirements baseline, repository structure, authentication module	Samarth	Working login, Git repo with initial commits
Sprint 2	Workspace and project creation, task/subtask CRUD, Kanban board	Sameeksha	Demo of Kanban board, unit tests
Sprint 3	Backlog creation, sprint planning, story point estimation	Sanjana	Demo of backlog to sprint flow, unit tests
Sprint 4	Task dependencies, time logging, burndown and velocity charts	Sanjana / Sameeksha	Integration tests, working charts
Sprint 5	Real time board updates, comments, mentions, notifications	Hemanth	Live demo across two sessions, tests
Sprint 6	Analytics dashboard, security hardening, validation and final report	All	Validation report, final working demo

10. Proposed Technology and Tools

Layer / Activity	Tool / Technology
Backend framework	Django REST Framework (Python)
Real time layer	Django Channels (WebSockets)
Frontend	React (JavaScript)
Database	PostgreSQL
Version control	Git and GitHub
Agile planning and bug tracking	GitHub Projects / Jira
CI/CD	Jenkins
Containerization	Docker

Layer / Activity	Tool / Technology
Testing	Pytest, Jest
Static code analysis	SonarQube
Design and diagramming	Draw.io

11. Agile Justification

The project will be executed in six sprints, each ending with a working, demonstrable increment. A backlog of user stories will be maintained in GitHub Projects from the first sprint onward, with tasks moved through the board as work progresses. Each sprint will include planning at the start and a short review at the end, so that progress and blockers are visible throughout the project rather than only at the final submission. Agile is well suited to this project because the four functional areas are largely independent, which allows each member to plan, build and demo their module every sprint while integrating with the rest of the system incrementally.